

2025-2026

Happy Hawks

24813

Portfolio

Team Members & Roles

Jose L.

**Programming &
Team Captain**

Leads team day-to-day operation. Is involved in design, building, business, drive and leads programming process.



Frank C.

Building Lead

Leads the assembly of the robot and building components. Is involved in design and drive.



Isaiah B.

**Business & Marketing
Lead**

Leads business & marketing.
Handles sponsorships, fundraising, and branding.
Runs social media and tracks the team budget.



Sean G.

Design Lead

Leads the robots planned assembly and construction, with a focus on designs 3D printed components. Is involved in building and drive.



Andrea S.

**Driving Lead & Co
Captain**

Leads the robot driving during matches and trains the driver team.



Johanna A.

Business & Marketing

Assists with organizing events, outreach, and promoting the teams image. Is involved in programming



Emilio S.

Building

Assists with building the robot and coaching the drive team.



Brisa M

Business & Marketing

Assists with organizing events, outreach, and promoting the teams image.



Team Members & Roles

Jordynn R.

**Business & Marketing
Lead:**

Handles sponsorships,
assists with organizing
events, and outreach.



Victoria G.

Builder:

Supports robot assembly
and contributing to
improving components.



Abel M

Builder:

Supports robot assembly
and construction.



Hennessy T.

**Director of
Development:**

Leads in fundraisers, and
excuting overall in
fundraisers



Ivet L.

Director of Design:

Leads the creative vision
and strategy, guiding design
teams to deliver cohesive,
user-centered experiences
that align with business
goals.



Bianca C.

Builder:

Supports robot assembly
and construction,
contributing to building
and improving components.



Ethan G.

Builder:

Supports robot assembly
and construction,
contributing to building
and improving components.



Nicola C.

Programmer:

Designs, writes, and
maintains code to build
software, solve problems,
and bring digital ideas to
life.



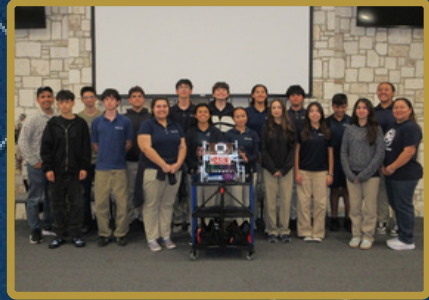
Team Overview & Future Vision & Expansion

Team Name:

24813

SHCA Happy Hawks

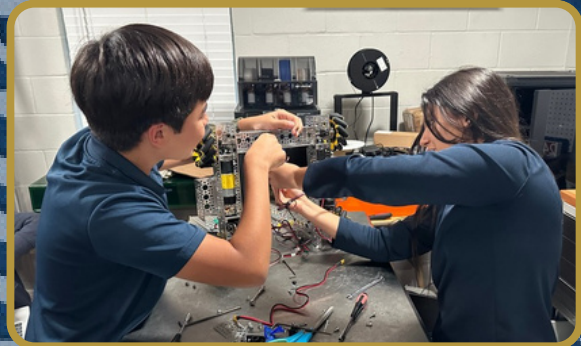
Season: DECODE



We are a student led robotics team focused on engineering, teamwork, and outreach. Our goal is to design a competitive robot while also growing leadership skills, communication, and community impact. Every member contributes to Building, Design, Programming, Business & Marketing, and Drive.

Mission and Values

Our mission is to create a positive, hardworking team environment where students learn real engineering skills and leadership. We value creativity, respect, and strong communication. We believe robotics should be both competitive and welcoming to all students.



Each member has a role so the team runs smoothly. Leadership focuses on communication and helping everyone feel included. We divide into sub-teams such as Building, Design, Programming, business/marketing, and Drive so work stays organized.

Future Goals

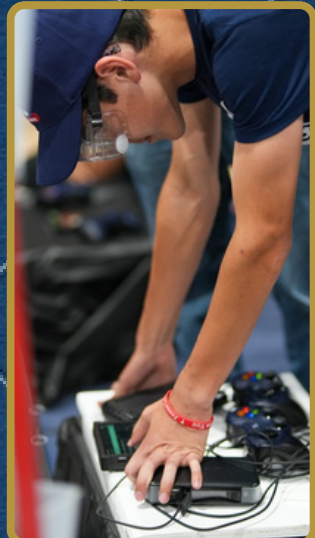
Moving forward, we want to: Grow sponsorships and outreach, Develop fully custom Robot, Utilize 3D printer to fullest extent, Mentor younger students, and develop a new FTC team under our wing.

Challenges and Growth

Throughout the season we faced challenges like time management and testing new mechanisms. Instead of giving up, we worked together to adjust designs and improve teamwork. These challenges helped us grow stronger as a team.

Season Highlights

- Hosted scrimmage
- Taught robotic classes in El Salvador
- Hosted robotic Vespers with NDAA
- Reached over 2,000+ people through Instagram
- Sponsored robotic parts to first Mexican-International Adventist robotics team



Outreach & Marketing

Sponsorships & Fundraising

Sponsorships played a major role in supporting our team throughout the season. We built strong, professional relationships with Laurel Ridge Treatment Center, R&L Overhead Doors, and Fred's Mobile Auto Services by clearly communicating our mission and demonstrating the impact of their support. Their contributions helped fund robot parts, competition expenses, and outreach opportunities.

In addition to sponsorships, we organized a variety of fundraising initiatives, including a GoFundMe campaign, bake sale, food fundraiser, and Fall Fest activities such as a maze and dunk tank. We also coordinated sponsor outreach and hosted events like our scrimmage. These efforts not only helped us meet our financial needs but also strengthened our connection with the community and increased awareness of our team.

Views ⓘ

2,404

Views

Followers 54.4%

Non-followers 45.6%

Accounts reached 465

By content type

All

Followers

Non-followers

Posts 74.0%

Stories 13.7%

Reels 12.4%

● Followers ● Non-followers



RoboCodex

1,128

Total subscribers

Social Media & Branding

Our business and marketing team focused on building a strong, consistent, and recognizable brand for the SHCA Happy Hawks. We created high-quality posts, graphics, and videos that showcased our progress, achievements, outreach efforts, and team culture while maintaining a cohesive visual style across all platforms.

Our strategy led to measurable growth and engagement. One of our videos reached over 3,300 views, and we grew to 85 followers, showing steady interest from our community. Overall, we reached thousands of people online, including over 60,000 YouTube views. This strong digital presence allowed us to promote our team effectively, attract support, and maintain meaningful connections with sponsors and supporters.

Outreach & Engagement

Impact on Community

Our team is dedicated to making STEM accessible and inspiring students of all ages and backgrounds, and the following is how we achieved that!

- Introduced robotics through engaging outreach events, demonstrations, and presentations
- Reached over 1,200 people at a Central Texas camp meeting
- Demonstrated robotics to over 500 people at our church
- Reached tens of thousands through social media
- Supported a one-week summer robotics camp hosted by our team captain, where students learned building and programming through hands-on activities
- Allowed students to interact with our robot, sparking interest in STEM
- Encouraged creativity, confidence, and problem solving skills
- Aimed to show students they can be part of the future of engineering and innovation

Fundraising Initiatives

To support our season, we organized multiple fundraising efforts.

- Created a GoFundMe campaign to raise funds
- Organized a bake sale that generated significant support
- Coordinated sponsor outreach and meetings
- Planned and hosted events like the scrimmage

These initiatives helped us reach our financial goals and ensured that our team could continue to compete and grow.

Initiatives & Events

This season, we actively promoted STEM through a variety of events and initiatives. One of our biggest accomplishments was hosting the first FTC Adventist Robotics Scrimmage, which provided teams with valuable practice, collaboration, and competition experience. We also participated in school and church events where we demonstrated our robot and allowed students to interact with it firsthand. These experiences made robotics more approachable and exciting, helping students see STEM as something they can be involved in.

Event Coordination & Leadership

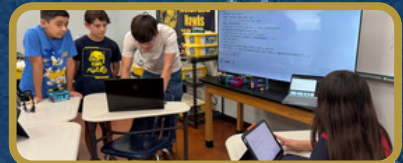
Our team took on a strong leadership role in planning and organizing events throughout the season. We were responsible for managing logistics, coordinating schedules, and ensuring that each event ran smoothly and efficiently.

For our scrimmage, we handled team registration, created match schedules, and managed the overall flow of the event. In addition, we mentored three robotics teams across different regions, providing guidance, support, and encouragement.

Through these leadership experiences, we developed important skills in organization, communication, and teamwork while creating meaningful opportunities for others to grow in STEM.

International Outreach – El Salvador/Mexico

Our impact extended beyond our local community through international outreach efforts. We worked with students in El Salvador, teaching LEGO robotics to K–12 students and hosting a one-week program focused on building, programming, and teamwork. We also supported a robotics team in Mexico by providing robot parts to help them continue developing and competing. Through these experiences, we shared knowledge, built connections, and inspired students to explore STEM on a global level.



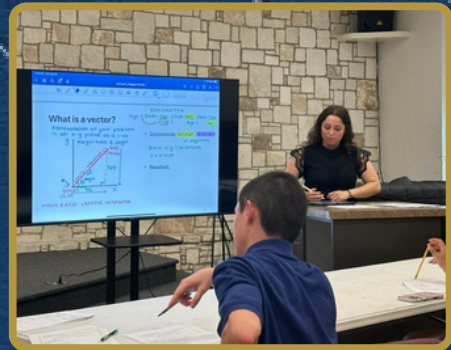
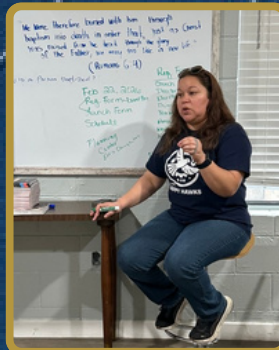
Mentors

Dr. Arturo Montoya

An Associate Professor at UTSA, Dr. Montoya helped launch our season with strong motivation and engineering insight. His guidance built our confidence and set a solid foundation for success.

Dr. Herta Montoya

An Assistant Professor in Mechanical, Aerospace, and Industrial Engineering, Professor Montoya guided us in applying mechanical design and control systems, helping turn our ideas into a competitive robot.



Mauricio Madrid

A Texas A&M Mechanical Engineering graduate, Mauricio supported the design of our shooter and intake systems. He taught us to think of our robot as integrated systems, improving reliability and performance.

Sonia Sifuentes

Our school's robotics and math teacher, Mrs. Sifuentes provided consistent support, guidance, and encouragement throughout the season.

Mrs. Acosta

Our business and marketing mentor, Mrs. Acosta guided our outreach, branding, and presentation efforts.

Letty Melero

As our fundraiser coordinator, Letty helped organize and support efforts to reach our team's financial goals.

Everardo Sifuentes

Everardo supported our programming efforts, helping improve our robot's functionality and performance.

Engineering Design Process

Robot Design Process

Observe: Find a Problem/Inconsistency or Bottleneck

Ask: Why does the Problem occur

Develop: A Hypothesis/Solution

Test: The Hypothesis/Solution

Analyze: The Results

Record: The Outcomes to Portfolio

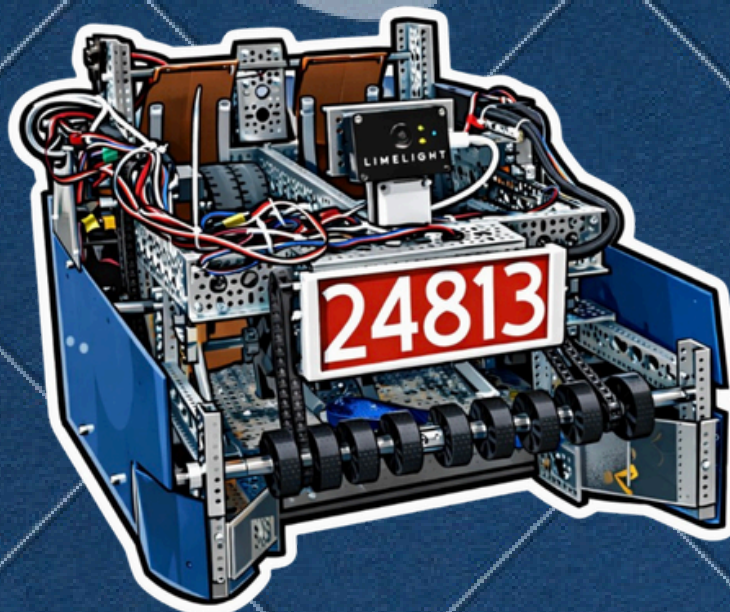
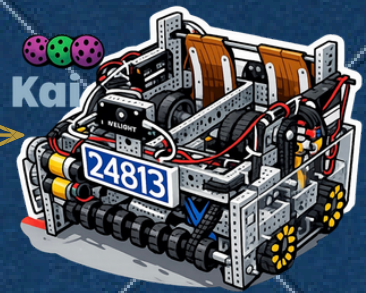
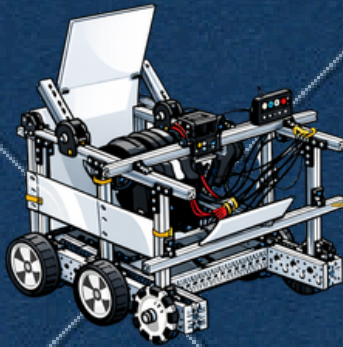
Conclusion: Decide whether to further improve

Rules to follow

All updates to the robot must:

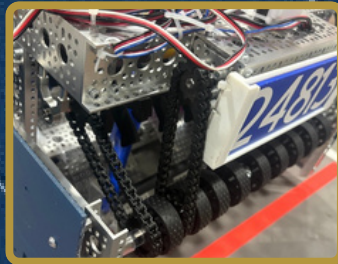
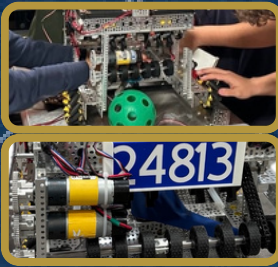
1. Quick Repair Access
2. Repeatable Autonomous Scoring
3. Durable Drive Train/Structure
4. Rapid Cycle Times
5. Reliable Shot Accuracy
6. Complies to FTC Rules/Standards

It is Vital to Communicate All Ideas Between The Designers, Programmers, and Builders to Insure a Clean Execution of Our Goals!!!



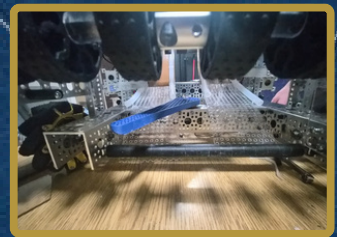
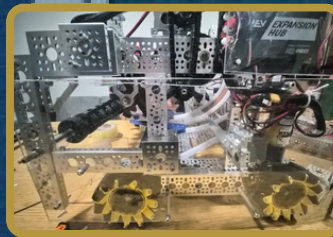
Intake development

The Intake system underwent multiple development iterations: Initially we struggled to achieve a balance of rotational speed and torque, while the feeder experienced problems with inconsistent compression throughout as well as dead spots. The diverter also had its own problems allowing the intake of 2 or 4 artifacts. The most significant limiting factor was the limited intake size of previous designs.



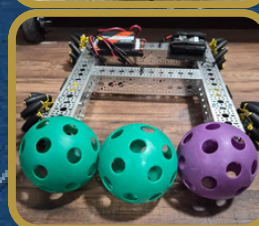
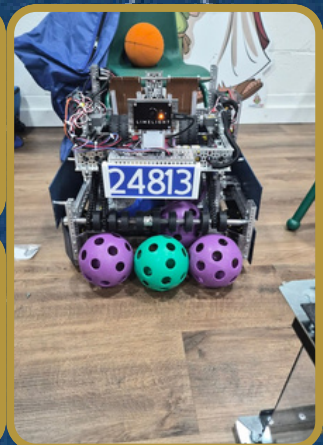
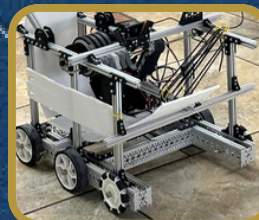
Our intake iterated from a single 6000 rpm motor, to a coupled 1150 rpm design, and finally a single 1150 motor with a 5:7 ratio motor to optimize speed to torque ratio.

The feeder required many minor optimizations to the boot kickers in order to perfect the artifacts compression with the guided hoods.



The diverter was designed to meet several key requirements. Its length was optimized to fit within the robot while still creating a specified angle for the guided shooter, and its width was optimized to eliminate detrimental dead zones without slowing intake speed. A lofted gradient was also added to direct forces up and toward an intended path. Achieving this required many 3D iterations.

When optimizing for cycle times, the size of the intake are critical. Our original REV-based design could only collect one artifact at a time, making it highly inefficient. The first iteration of our goBILDA-based design improved capacity to two artifacts but proved awkward in practice. In the final iteration, we implemented an overhang intake, significantly increasing efficiency by allowing the robot to collect multiple artifacts at once. This design widened the margin of error for drivers, resulting in faster and consistent cycle times.



Shooter Development

our hooded shooter has gone through 3 iterations and we have tried other shooting mechanisms such as a dual flywheel shooter and turret which we decided to scrap due to time constraints and limited materials. so we decided to optimize our existing shooter

1st iteration

Our first shooter was an iteration of the rev starter robot. The shooting was inconsistent, even trying to keep the flywheel consistent in the same rpm range was difficult. The balls would shoot out at seemingly random distances.

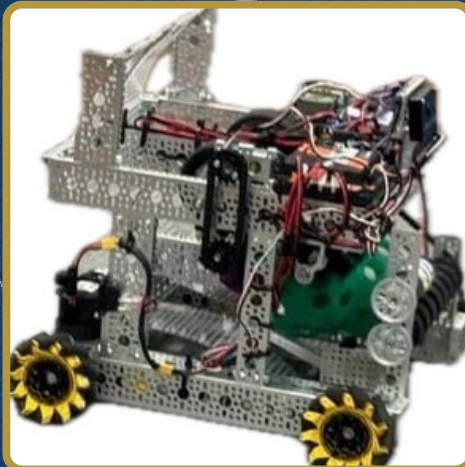


turret prototype

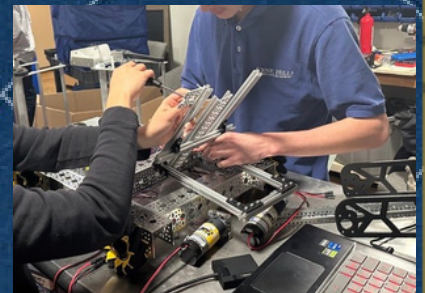


2nd iteration

We reengineered the shooter using the same core concept as the REV design, using polyurethane sheets chosen for their high impact resistance and ability to handle higher loads, and flexibility.

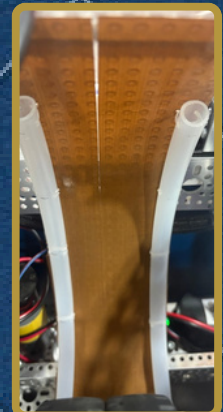


double flywheel prototype



3rd iteration

To further improve shooting accuracy, we reduced excess friction that was causing unwanted backspin and inconsistent bounces by applying Teflon tape to key contact surfaces, minimizing spin and improving bounce consistency. We also incorporated plastic tubing as rails, which helped guide the ball, enhancing overall consistency. Finally, we carefully tuned the system to balance flywheel speed in relation to the goal, resulting in a shooter that is reliable.



General optimization

Through testing and refinement, we identified several key issues. The robot would occasionally misfire when holding two to three balls, and uneven weight distribution affected its stability. We also encountered accuracy and consistency challenges, with shots frequently bouncing out of the goal. Additionally, we had concerns about durability, as exposed chains and gears left the robot vulnerable during frequent physical contact in matches.

Misfire?!

The misfiring issue was caused by the intake pushing balls into the shooter prematurely, so we implemented servo controlled stoppers to prevent this.

not long enough & let balls through



final iteration



100% infill

5mm width

not strong enough



10% infill

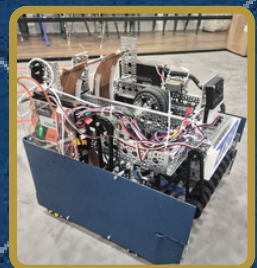
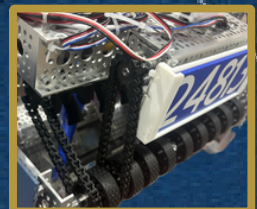
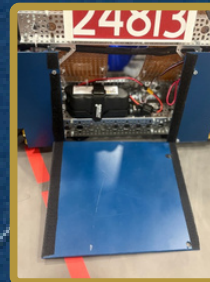
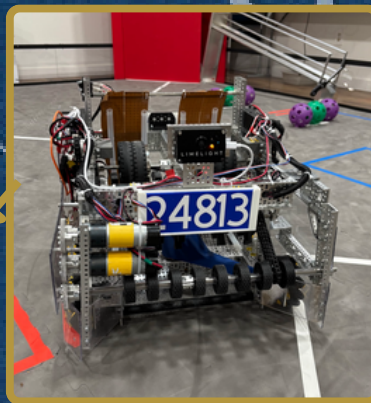
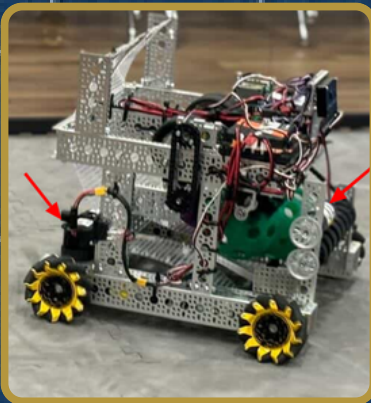
4mm width

Robot Armor

Concerns about our robot's safety led us to add a protective layer. We chose polycarbonate due to its high impact resistance. As a thermoplastic, it can be gradually heated and bent to the right shape.

Weight Distribution

The weight distribution issue was caused by having two motors and the battery positioned on the same side. So we moved the battery to the opposite side, removed one motor, and moved the remaining one to the middle of the robot.



Robot Gear Ratio Improvements

Subsystem	Actuator Used	Initial Ratio	New Ratio	Result
Drive Train	323 rpm motor	Same	Same	No change; kept proven balance.
Intake	1050 rpm motor	1:1	5:7	More torque, slower speed, less stalling.
Feeder	goBILDA Dual-Mode Torque Servo	1:1	7:5	Faster feeding, but lower torque.
Shooter	6000 rpm motors	1:1	2:3	More torque, lower speed, steadier under load.

Software: Motion Control & Pathing



PedroPathing is an advanced Reactive Vector Follower developed by FTC Team 10158. It uses Bézier curve generation to produce smooth, fast, and precise autonomous trajectories. Unlike time-based followers, PedroPathing is not speed-limited by its paths, allowing us to maximize our scoring and iteration within the 30-second autonomous window.

Why PedroPathing over Roadrunner

Through testing, we found PedroPathing was more precise at higher accelerations and better at correcting unexpected disturbances mid-path, which directly improves autonomous consistency.

FSM AUTO FLOW

IDLE → SPINUP → LAUNCH

Our autonomous routine uses a 4-state Finite State Machine. The robot begins IDLE, then skips using the Limelight 3A to ALIGN to the target since the robot can just go to a preset point, SPINs UP the shooter to the calculated RPM, then LAUNCHes. This structure ensures each action only begins when the previous is complete, improving reliability.

4 Autonomous Strategy

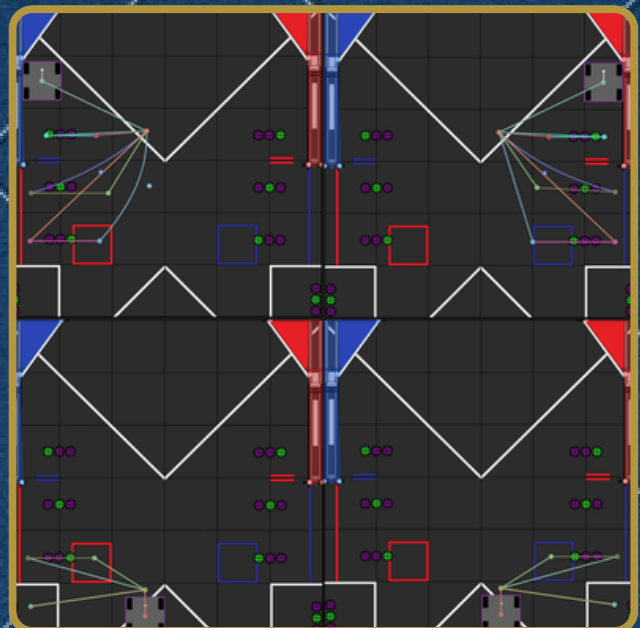
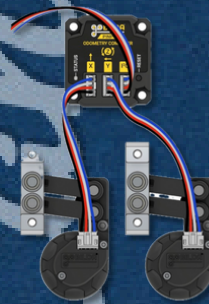
We developed four autonomous routines for different starting positions on the field, allowing our robot to adapt based on alliance strategy and partner positioning. These routines are split between starting near the GOAL and starting in the back LAUNCH ZONE, each optimized for efficient pathing, fast cycling, and consistent scoring. By using smooth trajectories and coordinated intake/shooting sequences, each autonomous has the potential to score up to 12 ARTIFACTS while minimizing downtime and maximizing reliability.

The Math Behind Reactive Vector Followers

Pedro Pathing combines an attractive force pulling the robot toward the goal with repulsive forces avoiding obstacles. Recalculating the ideal movement vector every loop cycle. Overall pedroPathing has so many good features its hard not to pick it.

GoBilda Pinpoint Odometry

We use the GoBilda Pinpoint odometry computer combined with PedroPathing's localizer for position tracking. The Pinpoint reads two perpendicular dead wheels to give highly accurate X/Y displacement, feeding PedroPathing's follower with real-time pose data.



Software: Software: Sensors & Vision

LIMELIGHT 3A – Three Roles



Shooter Alignment

The Limelight feeds heading error into a PD loop that rotates the robot until it is aligned with the target.

Distance → RPM Regression

Using the Limelight's θ_y (vertical angle), a linear regression formula calculates distance. A second regression then maps that distance to the required shooter RPM.

Autonomous Target Detection

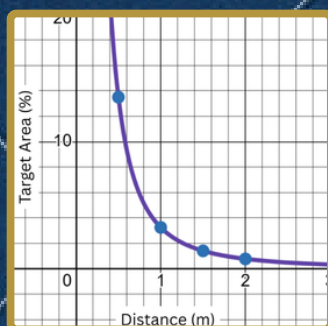
In autonomous, the Limelight detects the field artifact pattern to determine starting configuration and select the correct path.

Distance Estimation via Target Area Regression

The Limelight reports ta (target area as % of frame). As the robot moves closer, ta grows predictably. We can

reverse that relationship to calculate distance.

$AREA_SCALE$ and $AREA_EXPONENT$ were derived by collecting ta readings at known distances and fitting a power regression curve to the data.



```
// Distance to AprilTag using calibrated target area power model
//
// usages
//
public double getDistanceFromArea() {
    if (!hasTarget()) return 0.0;

    double ta = getTa();
    if (ta <= 0.0001) return 0.0;

    double distanceWithoutOffsetAdjustment = Math.pow((ta / AREA_SCALE), (1.0 / AREA_EXPONENT));
    double distanceWithOffsetAdjustment = distanceWithoutOffsetAdjustment + 25.5;
    // x = (ta / scale)^(1 / exponent)
    return distanceWithOffsetAdjustment;
}
```

GoBilda Pinpoint + Encoders

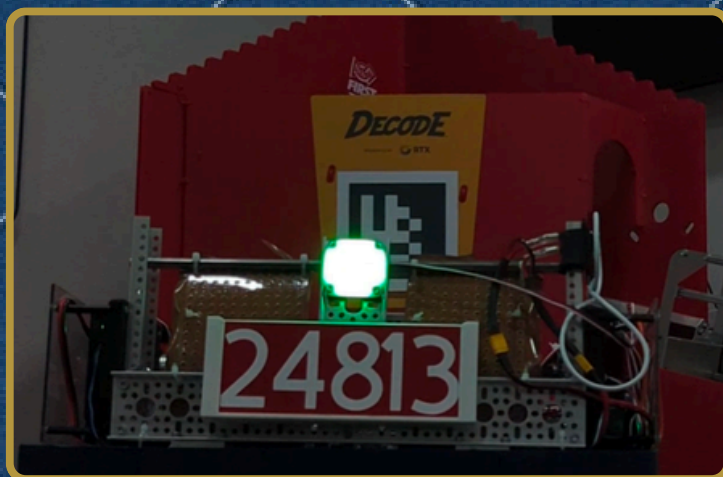
The Pinpoint odometry computer processes encoder ticks from two dead wheels to provide accurate field-relative positioning. Combined with the Limelight's relocalization capability, the robot maintains positional accuracy even after collisions or disturbances.

LED Alignment Indicator

A GoBilda PWM RGB LED gives drivers real-time alignment feedback. It turns **red** when the robot is outside the deadband and switches to **green** the moment alignment is achieved — eliminating guesswork during teleop.

PD Alignment Loop

The alignment system uses a PD (Proportional-Derivative) controller driven by the Limelight's horizontal angle (tx) as the error signal — tuned to $kP = 0.02$, $kD = 0.003$, deadband = 0.9° , and $maxPower = 0.4$. The robot continuously corrects its heading until tx falls within the deadband, at which point the FSM advances to SPINUP. Through comparative testing against a manual alignment approach assisted by RGB LED feedback, we found that experienced drivers could align faster manually. So for now the PD loop serves primarily as a fallback and consistency tool for newer drivers or high-pressure matches.



MISALIGNED LOCKED ✓
Controlled via servo PWM position mapping — $0.28 = \text{red}$, $0.50 = \text{green}$, linearly interpolated between.

Software: Shooter System

The Problem

RPM needed to vary with distance. Fixed RPM caused undershoot at far targets, overshoot at close ones. Instead we want the robot to be able to shoot from everywhere on the field.

Physics-Based Model - Research in Progress

Our programming lead began developing a physics-based shooter model as an alternative to empirical tuning. The model aimed to calculate required RPM from physics principles using projectile motion, flywheel inertia, and game piece aerodynamics rather than collected data points. While the approach was theoretically more generalizable, it was not implemented this season due to the complexity of tuning the physical constants accurately within our timeline.

The research paper is ongoing and available to view below.

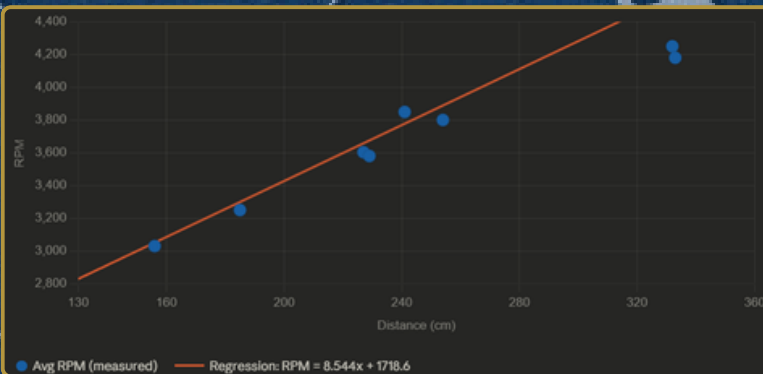
$$\text{RPM} = \frac{60}{2\pi r \eta} \cdot \left(\sqrt{\frac{g d^2}{2 \cos^2(\theta)(d \tan(\theta) - \Delta y)}} \right)$$

Physics Model Research Paper

"Physics-Based Modeling and Vision-Assisted Optimization of a Flywheel Shooter for the FTC DECODE™ Robotics Game" - In Progress, Jose Laurencio



Scan to view — this is a working draft, not a finalized paper.



DECODE RI3D — SHOOTER TUNING LOOKUP TABLE

Enter measured shot data below. Formulas auto-calculate averages and suggest ShooterModel constants.

#	DISTANCE		SHOOTER RPM TARGETS			ALIGNMENT OFFSETS (°)	
	Distance (cm)	Distance (in)	Left RPM Target	Right RPM Target	Avg RPM	Left Offset (°)	Right Offset (°)
1	185.0	72.8	3300	3200	3250	-3.000	6.500
2	227.0	72.8	3655	3550	3603	-4.500	
3	241.0	94.9	3849		3849	-4.500	
4	254.0	100.0	3800		3800	-2.400	
5	332.0	130.7	4250	4250	4250	-2.500	1.000
6	333.0	131.1		4180	4180		1.600
7	229.0	90.2		3580	3580		2.000
8	186.0	73.4		3030	3030		1.000

Shooter RPM Model

Once distance is calculated, a linear regression maps it directly to the required shooter RPM:

$$\text{RPM} = 8.54 \times \text{distance} + 1718.6$$

Constants were derived empirically by recording shooter RPM and measuring resulting shot accuracy at multiple distances. The model allows the robot to automatically set the correct flywheel speed from any field position — no manual adjustment needed between shots.

```
public static double distanceToRPM(double d, boolean isLeftShooter) {  
    if (USE_LINEAR_MODEL) {...}  
  
    // PHYSICS MODEL  
  
    double meters = d / 100.0;  
  
    double deltaY = targetHeight - shooterHeight;  
    double numerator = g * Math.pow(meters, 2);  
    double denominator = 2 * Math.pow(Math.cos(theta), 2)  
        * (meters * Math.tan(theta) - deltaY);  
  
    if (denominator <= 0) return 0;  
  
    double v0 = Math.sqrt(numerator / denominator);  
  
    double rimVelocityRequired = v0 * 2;  
  
    double rpm = (rimVelocityRequired * 60)  
        / (2 * Math.PI * wheelRadius * eta);  
  
    return rpm;  
}
```

Full Code & Equations

All regression formulas, PD constants, and shooter code are documented in our public GitHub repository. (QR code to GitHub on the next page)

Software: Architecture & Teleop

Finite State Machine Architecture

Our entire robot is controlled through a Finite State Machine. Rather than running commands in sequence, the FSM defines discrete robot states and the conditions required to transition between them. This makes the code modular, easy to debug, and reliable under match pressure.

Flow diagram (main teleop FSM):

IDLE → ALIGN → SPINUP → LAUNCH → (back to IDLE)

Each state has a clear entry condition, active behavior, and exit condition. No state can be skipped, preventing partial execution errors.

Why FSM over Linear Code

Linear OpMode	FSM
Steps run in order, blocking	States run in parallel, non-blocking
Hard to interrupt mid-action	Clean transitions at any point
Difficult to debug	Each state is isolated and testable

Teleop Highlights

- **Self-Aligning:** Driver holds a button → FSM transitions to ALIGN → Limelight PD loop corrects heading automatically → releases control back to driver once locked on
- **Auto-RPM:** Once aligned, shooter PIDF spins up to the regression-calculated RPM automatically — driver only needs to trigger launch
- **Stall Protection:** Intake motor monitors current draw; when a stall is detected, the motor cuts power automatically to prevent burnout (directly addresses the burned motor issue from earlier in the season)
- **Dual Launch Modes (Vision & Manual):** The robot supports two shooting modes toggled with the BACK button. In Vision Mode, the Limelight calculates distance and automatically sets the flywheel RPM. In Manual Mode, the driver aims freely while the robot still pulls a distance-based RPM snapshot from the Limelight at the moment of trigger — preserving accuracy without requiring alignment.
- **Dynamic Target Offset:** Rather than using a fixed alignment target, the robot calculates a distance-dependent horizontal offset for each shooter left or right ($\text{BASE_OFFSET} + \text{distance} \times \text{DISTANCE_FACTOR} \pm \text{SHOOTER_OFFSET}$), accounting for the physical separation between the two flywheels at varying ranges.
- **RGB Alignment Indicator:** A GoBilda PWM RGB LED linearly interpolates between red and green based on alignment error. It turns fully green only when tx falls within the deadband — giving drivers a clear visual lock confirmation without reading telemetry.
- **Manual Shot Data Logging:** During manual shots, the robot logs the exact velocity target via telemetry (Manual L/R target TPS). This allowed the team to collect RPM-at-distance data points at known field positions, which directly fed into building and validating the shooter regression model.

Public GitHub Repository

Our full codebase is publicly available on GitHub for other teams to reference, learn from, and build upon. It's also really important for our version control and documentation to see how our code changed throughout the season.

HappyHawks/ftc-decode on github.



Autonomous Score Progress

Max autonomous score: 29 pts | Avg improving toward consistent 20+
Continued development focused on faster cycles and shooter PID consistency

Milestone	Software Implementation	Resulting Impact
Pathing Upgrade	Switched from Linear OpMode to PedroPathing.	Faster, non-blocking cycles within the 30s window.
Vision Lock	Integrated Limelight 3A with a PD alignment loop.	Higher accuracy; eliminated missed shots due
Logic Maturity	Moved full control to a Finite State Machine (FSM).	Reliability; eliminated partial execution errors

Closing & Connect

Closing Statement

The SHCA Happy Hawks are committed to growing in STEM, strengthening our teamwork, and continuing to make a positive impact in our community and beyond. This season has allowed us to develop our technical skills, leadership, communication, and outreach efforts while inspiring others to explore robotics.

We are proud of the progress we have made and excited to continue building, learning, and sharing our passion for STEM in the future.



Thank You

Thank you to our sponsors, mentors, school, church community, and supporters for believing in our team. Your encouragement and support make it possible for us to compete, grow, and reach more students each season.

We truly appreciate everyone who has contributed to our journey.



Connect With Us

Learn more about our team on our website:



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Stay connected to see our progress, outreach, events, and future updates!

